

Module -8: Network Access Basic Routing and Advance Routing concept, Switching concepts

1) Explain Switch

Answer:

- A switch is a networking device used to connect multiple devices in a LAN network.
- It works at Layer 2 of the OSI model and forwards data using MAC addresses.
- A switch sends data only to the destination device, improving speed and efficiency.
- It reduces network traffic and improves communication performance.

2) Explain Switch Boot Sequence

Answer:

- The switch boot sequence is the process that starts when a switch is powered on.
- First, the switch performs POST (Power-On Self-Test) to check hardware components
- Then it loads the boot loader and Cisco IOS from flash memory.
- Finally, the startup configuration file is loaded into RAM and the switch becomes operational.

3) Explain Three Methods to Access Switch Command Line Interface

Answer:

1. Console Access

- **Direct connection using a console cable.**
- **Used for initial configuration and troubleshooting.**

2. Telnet Access

- **Remote access over a network using IP address.**
- **Not secure because data is sent in plain text.**

3. SSH (Secure Shell)

- **Secure remote access method with encryption.**
- **Commonly used for managing network devices securely.**

4) Explain and Configuring the Cisco Internet Operating System (IOS)

Answer:

- **Cisco IOS is the operating system used in Cisco routers and switches.**
- **It helps configure, manage, and monitor network devices using CLI commands.**
- **Basic configuration includes setting hostname, passwords, IP addresses, and saving configuration.**
- **Commands like enable, configure terminal, and show running-config are commonly used in IOS.**

5) Explain Switch Port

Answer:

- **A switch port is a physical interface on a switch used to connect devices like PCs, printers, or routers.**
- **Each port has its own bandwidth and MAC address learning capability.**
- **Switch ports can operate in access mode or trunk mode.**
Ports help devices communicate within the network efficiently.

1) R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

A. R1

B. R2

C. R3

D. R4

E. None of the routers will exchange routing information.

Answer:

A. R1

B. R2

R1 and R2 are using the same OSPF process, so they can form neighbor relationships and exchange routing information.

2) enable secret [password] is hashed using which algorithm?

A. MD5

B. AH

C. PSK

D. ESP

E. WPA2

Answer:

A. MD5

3) An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

- A. R1 is an Area Border Router.**
- B. R1 is a backup designated router.**
- C. Router 2.2.2.2 is an Area Border Router.**
- D. Router 2.2.2.2 is a backup designated router**

Answer:

- D. Router 2.2.2.2 is a backup designated router.**

4) Which command is used to view the neighbor discovery table on a PC?

- A. show ipv6 neighbor**
- B. show ipv6 neighbors**
- C. netsh interface ipv6 show neighbor**
- D. netsh interface ipv6 show neighbors**

Answer:

- C. netsh interface ipv6 show neighbor**

5) What type of variable is being shown? Routers = [R1,R2,R3]

- A. List**
- B. Dictionary**
- C. Simple**
- D. Unsigned integers**

Answer:

- A. List**

6) Identify the fields in an IPv4 header. (Choose three)

A. Host component

B. Time to Live

C. Source address

D. Destination address

E. Network address

Answer:

- **B. Time to Live**
- **C. Source address**
- **D. Destination address**